AI 101 for Mechanical Engineers · Module 16

Submit evidence, not only a browser score

Lesson 16.3

What you will learn

> Distinguish the browser knowledge grade from practical assessment, explain the mini-project rubric, and prepare work another person can rerun, review and assess within stated operating limits.

Lesson overview

Module 16 · Lesson 3

Submit evidence, not only a browser score

AI 101 for Mechanical Engineers

1 The browser score checks knowledge

Best module quiz scores, mean: 70% | **Final: 30%**

Completion record: at least 70% on every module quiz and on the final. Lessons and flashcards do not count.

- It cannot authenticate a learner
- It cannot validate a model
- It cannot authorize deployment

The page contains the answer keys and progress can be edited.

2 The practical rubric weighs evidence

Data and preprocessing 45% | **Evaluation and physics 45%** | **10%**

Data and preprocessing

Problem and data	**20%**
Baseline, split and preprocessing	**25%**

Evaluation and physics

Evaluation and interpretation	**25%**
Physical verification and limitations	**20%**

Reproducibility and AI-use disclosure

Runs again; AI help disclosed	**10%**

An instructor or peer reviews the notebook; the course cannot certify it automatically.

3 Before you submit

- Restart the kernel and run every cell from top to bottom
- Fix hidden variable dependencies; inspect outputs
- State the operating domain: machine bounds, operating parameters, load ranges
- Include independent physics checks and AI disclosure
- Explain one modelling choice and one failure mode

A completion record is not evidence that a model is safe or ready to deploy.

Key takeaway

The browser score shows knowledge; the project shows engineering evidence.

Submit work a reviewer can rerun, with its limits stated.

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The browser grade combines quizzes and final

Component	Weight
Mean of best module quiz scores	70%
Final blueprint assessment	30%
Lessons and flashcard ratings	Not counted

> Optional completion record: at least 70% on every module quiz and the final.

A browser score is only a knowledge check

- It is not a practical evaluation
- Answer keys are exposed and progress can be edited
- It cannot authenticate a learner or establish model validity
- It cannot authorize deployment

The practical rubric weights engineering evidence

Group	Detailed breakdown
Data and preprocessing (45%)	Problem/data 20% + baseline/split/preprocessing 25%
Evaluation and physics (45%)	Evaluation/interpretation 25% + physical verification/limitations 20%
Reproducibility and AI disclosure (10%)	Reproducibility/AI-use disclosure 10%

> The same three groups, shown at two levels of detail.

Questions the rubric asks

1. Are the problem and data well defined?
2. Are the baseline and split justified?
3. Is preprocessing isolated from evaluation data?
4. Are results interpreted with physical checks?
5. Is the work reproducible, with appropriate AI-use disclosure?

People, not the course, review the notebook

- The course cannot inspect or certify a notebook automatically
- An instructor or peer reviewer can inspect the evidence
- They can ask you to reproduce a result

Make the submission reproducible and bounded

1. Restart the kernel; run every cell top to bottom
2. Resolve unrecorded variable dependencies and inspect outputs
3. State model-selection rationale, failure modes or thresholds
4. State the operating domain: machine bounds, parameters, load ranges

Submit evidence a reviewer can rerun

- Runnable notebooks, domain limitations, independent physics checks, AI disclosures
- Preserve data, environment and decision information
- Explain one modelling choice and one failure mode
- Never present a completion record as evidence of safety or readiness

CHECK YOUR UNDERSTANDING

What can a clean, reproducible project show that a browser score cannot?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What can a clean, reproducible project show that a browser score cannot?

> It runs top to bottom, justifies split and baseline, reports errors with units and physical checks, states limits, and can be rerun. A browser score only checks knowledge.

CHECK YOUR UNDERSTANDING

Which operating limits must a reviewer see?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which operating limits must a reviewer see?

> The physical operating domain: the valid machine bounds, operating parameters and load ranges for which the results are claimed.

Key takeaways

- Browser grade: 70% mean best quiz score, 30% final assessment
- Browser scores cannot authenticate learners, validate models or authorize deployment
- Rubric: data/preprocessing 45%, evaluation/physics 45%, reproducibility/AI disclosure 10%
- Restart the kernel and run the notebook top to bottom
- Submit runnable notebooks, operating limits, physics checks and AI disclosures